Proof of the Continuum Hypothesis

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Abstract

German mathematician Cantor proposed the concept of equipotence for two infinite sets: if a one-to-one mapping can be constructed between the elements of two infinite sets, they are called equipotent. Why can one-to-one mapping distinguish the power of sets? This article provides a clear explanation using computational methods. There are two types of one-to-one mapping: partial field bijective mapping and full field bijective mapping. If these two mappings are not distinguished, it is easy to lead to fallacies when using the method of proof by contradiction to prove non-equipotence. It has been proven through computational methods that real numbers and natural numbers are not equipotent. This also yields a method for proving irrationality. Finally, the continuum hypothesis was discussed.

keywords: partial field bijective mapping, full field bijective mapping, continuum hypothesis, irrationality
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1 Computation of Set

German mathematician Cantor introduced the concept of equipotence for two infinite sets[1, 28]: If a one-to-one mapping can be constructed between the elements

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of two infinite sets, then these two sets are said to be equipotent, meaning their cardinalities are the same. Why can a one-to-one mapping be constructed between equipotent sets? What is the fundamental reason? This article will provide a clear explanation. To compare the power and number of elements of two sets, we can directly perform the calculation. Use (A) to represent the number of elements in set A, and $|A|$ to denote the cardinality of the set A.

To effectively distinguish the sizes of infinite sets, an isometric mapping is introduced. For any two source elements A_1 and A_2, if their mapped image elements are B_1 and B_2, respectively, and the source distance equals the image distance, then this is called an isometric mapping, denoted by

$$|A_1 - A_2| = |B_1 - B_2| .$$

Assuming $(A) = N$, (B) is a function of N, denoted by $f(N)$, such as $N/2$, N^2, N^N, etc. If the function $f(N)$ has an upper bound, we can say that the set is countable, and the two are equipotent, i.e. $|A| = |B|$. We can use N as the unit (independent variable) to express (B) by a finite function. Finite functions can be represented by $f_L(N)$. Thus,

$$(B) = f_L(N) = N^{N^{N^{\cdot^{\cdot^{\cdot^{N}}}}}} , \text{ finite } N\text{s}.$$

If $f(N)$ has no upper bound, it cannot be expressed by any finite expression, and can only take the form of $f(N) = N^{N^{N^{\cdot^{\cdot^{\cdot^{N}}}}}}$, there are infinite Ns, where the infinity is independent of N, e.g., $f(2) = 2^{2^{2^{\cdot^{\cdot^{\cdot^{2}}}}}}$, then this set is uncountable, and the two are not equipotent, i.e. $|A| \neq |B|$. Using N as the unit (independent variable), no finite function expression can be found. Use $f_U(N)$ to represent an infinite function.

$$(B) = f_U(N) = N^{N^{N^{\cdot^{\cdot^{\cdot^{N}}}}}} , \text{ infinite } N\text{s}.$$

So how to find the function $f(N)$?

For two infinite sets A and B, if $(A) = N$, and under an equidistant mapping, each element A can correspond to X elements B_k, then $(B) \geq NX$. If X has an upper bound, which is Y, then

$$(B) \leq NY, \ B \text{ is countable.}$$

If X has no upper bound, then B is uncountable,

$$(B) = f(N) = N^{N^{N^{\cdot^{\cdot^{\cdot^{N}}}}}} , \text{ infinite } N\text{s}.$$

Use this method to calculate common sets.

Assume

$$(\{\text{Natural Numbers}\}) = N,$$

$$(\{0 < x \leq 1\}) = R.$$

Calculation results:

$$\big(\{x \in R^+ \mid x \leq N\}\big) = NR.$$

Under an equidistant mapping, each even number corresponds to two natural numbers: $2k$ corresponds to $2k-1$ and $2k$. Therefore, the number of elements in the set of natural numbers (N) is twice the number of elements in the set of even numbers (EN). We have:

$$(EN) = f(N) = \frac{N}{2}.$$

Equidistant mapping is not the only method for computation; according to the definition of a set, any method consistent with that definition may be employed. For instance, the enumeration of the set of rational numbers can be carried out using combinatorial methods.

$$2N < (Q^+) = f(N) < N^2,$$

$$\text{so: } |Q^+| = |N|.$$

2 PFBM and FFBM

The one-to-one mapping constructed by Cantor is a partial field bijective mapping (PFBM), in which bijection exists between only partial fields of two sets, not a full field bijective mapping (FFBM) [2]..

The one-to-one mapping between even numbers (EN) and natural numbers (N) is a PFBM, not an FFBM. PFBM can borrow numbers from outside the field, but when it comes to the FFBM, there is nowhere to borrow from.

If $(A) \neq (B)$, there will be no FFBM between A and B, and only PFBM is possible. For example, the bijection between EN and N is as shown in table 1.

N	1	2	3	4	...
EN	2	4(Borrow)	6(Borrow)	8(Borrow)	...

Table 1. PFBM between N and EN

Consider the difference of $f(N)$ at point k: $\Delta f(k) = f(k+1) - f(k)$ is a function with finite values (e.g. $2k$, k^k, etc.). The calculated value is a natural number, and the set of natural numbers (or its subset) can borrow numbers from behind. As shown in Figure 1.

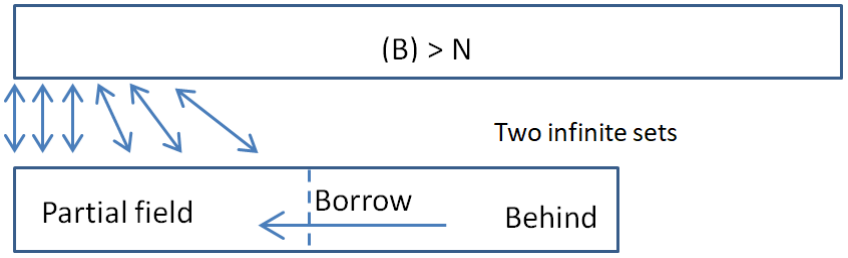


Figure 1 PFBM

However, if the difference $\Delta f(k)$ is an infinite function, there is not enough numbers to borrow, and a PFBM cannot be formed.

$|A|$ denotes the cardinality of the set A, and the following relationship can be derived, as shown in Figure 2.

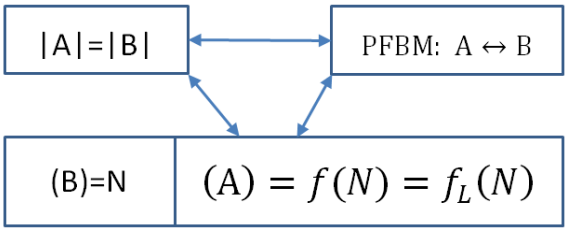


Figure 2 Equipotence

These three propositions are equivalent. This is the fundamental reason for the validity of the Cantor-Bernstein theorem. Similarly, the negations of these three propositions are also equivalent, As shown in Figure 3.

We have Lemma 1: $|N| = |f_L(N)|$, where the difference of $f_L(N)$ is finite.

Cantor's theorem[3, 47–50] states that the power set of a countable infinite set is uncountable. However, based on the previous reasoning, the natural numbers can form a PFBM with its power set as shown in table 2.

If $X \in N$ and $X \leq N/2$, then $2X \in N$; a PFBM is constructed.

Why is there a contradiction? It's because the method of proof by contradiction assumes that two sets can construct a one-to-one mapping, and the assumed one-to-one mapping is an FFBM. The result of the proof is that an FFBM cannot be

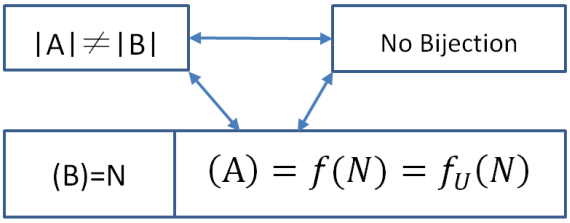


Figure 3 Non-equipotence

Value of N	1	2	3	preceding term +1	...	+1
Power set 2^N	2	4	8	preceding term ×2	...	×2
N	2	4	8	preceding term ×2	...	×2

Table 2. PFBM between N and its power set

constructed, but it cannot be denied that a PFBM can be constructed, As shown in Figure 4.

Cantor did not distinguish the difference between the two mappings, which led to

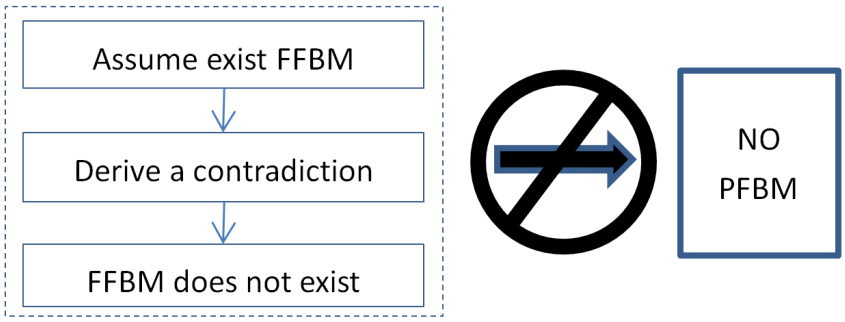


Figure 4 Proof by contradiction

the mistake of considering power sets as uncountable. Using the same method of proof by contradiction, it can be proven that there cannot be an FFBM between the set of N and EN. A PFBM can be constructed even between N^N and N, since N^N has finite representation.

3 Proof of Cardinality Inequality Between N and R

Next, we will prove that the number of real numbers in (0,1) (R) is uncountable with N as the base. That is,

$$R = f_U(N) = N^{N^{N^{\cdot^{\cdot^{\cdot^{N}}}}}}, \quad \text{infinite } N\text{s}.$$

The set of natural numbers is divided into two parts: one part consists of those that can be expressed by numerical expressions, referred to as expressible natural numbers; the other part consists of those that cannot be expressed by numerical expressions, referred to as inexpressible natural numbers, as shown below:

$$\underbrace{1 \quad 2 \quad 3 \quad 4 \quad 5 \quad 6 \quad \cdots}_{\text{expressible natural numbers}} \underbrace{\cdots N-3 \quad N-2 \quad N-1 \quad N}_{\text{inexpressible natural numbers}}$$

The sum or product of two expressible natural numbers is still an expressible natural number; that is, the set of expressible natural numbers is closed under addition and multiplication. However, the sum or product of two natural numbers may not be a natural number. For example,

$$(N-1)+3 = N+2 > N, \quad N \times 2 = 2N > N, \quad \text{and} \quad N \times N = N^2 > N.$$

For any expressible natural number m, $m < N \Rightarrow m^2 < N \Rightarrow m < N/m$. The larger m is, the smaller $1/m$ becomes. Therefore, the proportion of expressible natural numbers among all natural numbers tends to zero.

According to the definition of rational numbers, a real number $0 < r < 1$ is called a rational number if there exist natural numbers p and q, such that:

$$r = \frac{p}{q}.$$

Given the existence of inexpressible natural numbers, the sum or product of two rational numbers may not be a rational number, but rather an irrational number. For example,

$$\frac{1}{N} \times \frac{1}{3} = \frac{1}{3N};$$

the result is not a rational number because $3N > N$ and thus is not a natural number.

For every natural number, there exists a reciprocal in the interval $(0, 1]$; therefore, $R \geq N$. Each natural number k corresponds to $1/k$, and $1/N$ is a rational number.

Since $1/N \times 1/N = 1/N^2$ and $1/N^2$ is an irrational number, considering the sequence

$$0 < \frac{1}{N^2}, \frac{2}{N^2}, \frac{3}{N^2}, \ldots, \frac{N^2-2}{N^2}, \frac{N^2-1}{N^2}, \frac{N^2}{N^2} \leq 1,$$

we have $R \geq N^2$.

Similarly, since $1/N^2 \times 1/N^2 = 1/N^4$ and $1/N^4$ is an irrational number, considering the sequence

$$0 < \frac{1}{N^4}, \frac{2}{N^4}, \frac{3}{N^4}, \ldots, \frac{N^4-2}{N^4}, \frac{N^4-1}{N^4}, \frac{N^4}{N^4} \leq 1,$$

we have $R \geq N^4$.

By continuing the calculation in the same manner, we find that the exponent of N does not stop at N. Thus, we obtain:

$$R \geq f_L(N) = N^{N^{N^{\cdot^{\cdot^{\cdot^N}}}}}, \quad \text{finite } N\text{s}.$$

Then we have

$$R = f_U(N) = N^{N^{N^{\cdot^{\cdot^{\cdot^N}}}}}, \quad \text{infinite } N\text{s}.$$

Therefore, R is uncountable with N as its base, $|R| > |N|$. There is no partial field one-to-one mapping between the two sets.

If $R = f_L(N) = 2^N$, then the continuity of functions such as $f(x) = x^2$ cannot be guaranteed. In this case, the smallest number is $1/R$, and we have:

$$\frac{1}{R} = \frac{1}{f_L(N)} = \frac{1}{2^N}.$$

To guarantee the continuity of $f(x) = x^2$, there must exist $r \in R$ such that $f(r) = r^2 = 2$, where $1 < r < 2$. For the number $2 + \frac{1}{2^N}$, there must exist $m \in R$ such that:

$$\begin{aligned} f(r+m) = (r+m)^2 &= 2 + \frac{1}{2^N} \\ &\Rightarrow r^2 + 2rm + m^2 = 2 + \frac{1}{2^N} \\ &\Rightarrow 2rm + m^2 = \frac{1}{2^N} \\ &\Rightarrow 2rm < \frac{1}{2^N} \\ &\Rightarrow m < \frac{1}{r \cdot 2^{N+1}} < \frac{1}{2^N} = \frac{1}{R}. \end{aligned}$$

To ensure the continuity of the function, m must be smaller than the smallest number; that is, $R \neq f_L(N) = 2^N$.

Consider the Wallis Product:

$$\frac{\pi}{2} = \prod_{n=1}^{\infty} \frac{4n^2}{4n^2 - 1} = \prod_{n=1}^{M} \left(\frac{2n}{2n-1} \cdot \frac{2n}{2n+1} \right).$$

Here, the largest natural number N involved is either $2M + 1$ or $2M + 2$. The expanded form is:

$$\frac{\pi}{2} = \left(\frac{2}{1} \cdot \frac{2}{3} \right) \times \left(\frac{4}{3} \cdot \frac{4}{5} \right) \times \left(\frac{6}{5} \cdot \frac{6}{7} \right) \times \cdots$$

If both the numerator and the denominator were natural numbers, then $\pi/2$ would be a rational number. However, since $\pi/2$ has been proven to be irrational, at least one of the numerator or denominator must not be a natural number. Given that the denominators are all odd numbers, the factor of 2 in the numerator cannot be canceled out after simplification. Thus, we have:

$$\frac{\pi}{2} = \frac{2^{2M} \cdot p}{q}, \quad p \geq 1 \text{ and } q > 1.$$

Since $2^{2M} \cdot p > 2M + 2$, the numerator cannot be a natural number. Therefore, $\pi/2$ is irrational.

This corroborates the validity of the logic above and can also serve as a method for proving irrationality.

4 Regarding the Continuum Hypothesis

The set of real numbers is commonly referred to as the continuum. Does there exist an infinite set whose power is greater than that of the set of natural numbers but smaller than that of the set of real numbers? This question is known as the continuum hypothesis[4, 97–99].

$f(N)$ is either finite or infinite, i.e., $|f(N)| = |N|$ or $|f(N)| = |R|$.

Based on the concept of partial field bijective mapping, we have six lemmas:

Lemma 1: $|N| = |f_L(N)|$, where the difference of $f_L(N)$ is finite.

Lemma 2: When $(A) > (B)$, it suffices to assign to each element of A a corresponding element in B. If there exists a partial injection from A to B ($f_p : A \to B$), then

$|A| = |B|$. That is

$$\left.\begin{array}{c}(A) > (B)\\ f_p : A \to B\end{array}\right\} \Rightarrow |A| = |B|.$$

Lemma 3: Given three infinite sets A, B, C, if $(A) > (B) > (C)$ and $|A| = |C|$, then

$$|A| = |B| = |C|.$$

Lemma 4: If $|A| = |B|$, we have $|f((A))| = |f((B))|$, where $f((X))$ is an unbounded function.

Lemma 5: Given four infinite sets A, B, C, D, if there exist partial injections $f_p : A \to B$ and $f_p : C \to D$, then a partial injection $f_p : A * C \to B * D$ must also exist.

Lemma 6: If $|A| = |B|$ and $|C| = |D|$, then we have $|A * C| = |B * D|$.

These lemmas will be used repeatedly in the subsequent calculations.

Since $|N| = |N^2| \Rightarrow |\sqrt{N}| = |N| \Rightarrow \left|\sqrt{N}^{N^{N^{\cdots^N}}}\right| = \left|N^{N^{N^{\cdots^N}}}\right| \Rightarrow |\sqrt{R}| = |R|$, we have $|R| = |\sqrt[2^k]{R}, k \in N|$.

Therefore $|f_L(N)| = |N| < |R| = |\sqrt{R}| \Rightarrow f_L(N) < \sqrt{R}$

$$\Rightarrow |R * f_L(N)| = |\sqrt{R}\sqrt{R}| = |R|.$$

We obtain

$$\frac{R}{f_L(N)} > \sqrt{R} \Rightarrow |\sqrt{R}| = \left|\frac{R}{f_L(N)}\right| = |R|.$$

It is easy to see

$$R > \frac{R}{2^k} > \sqrt{R} \Rightarrow |R| = \left|\frac{R}{2}\right| = \left|\frac{R}{2^k}\right|, k \in N.$$

We have

$$f_L(N) < \frac{R}{2} \Rightarrow |R \pm f_L(N)| = |R|.$$

The same holds when taking logarithms:

$$\left|\log_{f_L(N)} R\right| = |R|$$

Let $R = N^{N^{N^{\cdots^{N}}}}$, ∞ Ns, we have

$$\log_k R = N^{N^{\cdots^{N}}} \log_k N,\ \infty - 1\ N\text{s}.$$

Since $|N| = |N^N| \Rightarrow |N^N| = \left|N^{N^N}\right| \Rightarrow \cdots \Rightarrow |R| = \left|N^{N^{\cdot^{\cdot^{\cdot^{N}}}}},\ \infty - 1\ \text{Ns}\right|$

$$\Rightarrow |R| = |\log_N R|, \text{ and } (R) > (\log_k R) > (\log_N R),$$

we obtain $|R| = |\log_k R, k \in N| = |\log_N R| = |\sqrt{\log_N R}|$.
And so on, $|N| = |N^{N^N}| \Rightarrow |R| = |\log_N R| = \left|\log_N^{\log_N R}\right|$.
Let $f_L(N) = N^{N^{\cdots^{N}}}$, k $(k \in N)$ Ns, we have

$$\log_{f_L(N)} R = \frac{\log_N R}{\log_N f_L(N)} > \sqrt{\log_N R}.$$

Then, in the same way as before, we obtain $\left|\log_{f_L(N)} R\right| = |R|$.
Since $|\log_N R^{f_L(N)}| = |f_L(N) \log_N R| = \left|\sqrt{\log_N R} * \sqrt{\log_N R}\right| = |\log_N R|$, we have

$$|R^{f_L(N)}| = |R|.$$

In the same way, ${}^{f_L(N)}\sqrt{R} = R^{\frac{1}{f_L(N)}} \Rightarrow \log_N R^{\frac{1}{f_L(N)}} = \frac{\log_N R}{f_L(N)} > \sqrt{\log_N R}$

$$\Rightarrow R > {}^{f_L(N)}\sqrt{R} > \sqrt{\log_N R}$$

$$\Rightarrow \left|{}^{f_L(N)}\sqrt{R}\right| = |R|$$

Since $|\log_N [f_L(N)]^R| = |R * \log_N f_L(N)| = |\sqrt{\log_N R} * \sqrt{\log_N R}| = |\log_N R|$, we have

$$|[f_L(N)]^R| = |R|.$$

Based on R, the set resulting from addition, subtraction, multiplication, division, exponentiation, root extraction, and logarithms is still equipotent to R.

Let $T = F[R, f_L(N)]$, where F is one of the seven operations mentioned above, we have $|T| = |R|$. Then

$$|T| = |R| = \left|\frac{R}{2}\right| = \left|\frac{T}{2}\right| = |2R| = |2T|$$

$$= |\sqrt{R}| = |\sqrt{T}| = |\log_N R| = |\log_N T|$$
$$= |\sqrt{\log_N R}| = |\sqrt{\log_N T}|.$$

Therefore, we have

$$|F[T, f_L(N)]| = |R|.$$

The calculation process is as follows:
If $T \pm f_L(N) > R$, $2T > T \pm f_L(N) > R \Rightarrow |T \pm f_L(N)| = |R|$.
If $T \pm f_L(N) < R$, $\frac{T}{2} < T \pm f_L(N) < R \Rightarrow |T \pm f_L(N)| = |R|$.
If $T * f_L(N) > R$, $|T * f_L(N)| = |\sqrt{R}\sqrt{R}| = |R|$.
If $T * f_L(N) < R$, $T < T * f_L(N) < R \Rightarrow |T * f_L(N)| = |R|$.
If $\frac{T}{f_L(N)} > R$, $T > \frac{T}{f_L(N)} > R \Rightarrow \left|\frac{T}{f_L(N)}\right| = |R|$.
If $\frac{T}{f_L(N)} < R$, $\sqrt{T} < \frac{T}{f_L(N)} < R \Rightarrow \left|\frac{T}{f_L(N)}\right| = |R|$.
If $T^{f_L(N)} < R$, $T < T^{f_L(N)} < R \Rightarrow |T^{f_L(N)}| = |R|$.
If $T^{f_L(N)} > R$, $|\log_N T^{f_L(N)}| = |f_L(N) * \log_N T|$

$$= |\sqrt{\log_N R} * \log_N R| = |\sqrt{\log_N R} * \sqrt{\log_N R}| = |\log_N R|$$
$$\Rightarrow |T^{f_L(N)}| = |R|.$$

If $\sqrt[f_L(N)]{T} > R$, $T > \sqrt[f_L(N)]{T} > R \Rightarrow \left|\sqrt[f_L(N)]{T}\right| = |R|$.
If $\sqrt[f_L(N)]{T} < R$, $\log_N \sqrt[f_L(N)]{T} = \frac{\log_N T}{f_L(N)} < \log_N R$

$$\Rightarrow \log_N T < f_L(N) * \log_N R$$
$$\Rightarrow |\sqrt{\log_N T} * \sqrt{\log_N T}| = |f_L(N) * \log_N R|$$
$$\Rightarrow |\log_N \sqrt[f_L(N)]{T}| = |\log_N R|$$
$$\Rightarrow \left|\sqrt[f_L(N)]{T}\right| = |R|.$$

If $f_L(N)^T < R$, $T < f_L(N)^T < R \Rightarrow |f_L(N)^T| = |R|$.
If $f_L(N)^T > R$, $|\log_N f_L(N)^T| = |T * \log_N f_L(N)|$

$$= |\sqrt{\log_N R} * \sqrt{\log_N R}| = |\log_N R|$$
$$\Rightarrow |T^{f_L(N)}| = |R|.$$

If $\log_{f_L(N)} T > R$, $T > \log_{f_L(N)} T > R \Rightarrow |\log_{f_L(N)} T| = |R|$.

If $\log_{f_L(N)} T < R$, $\log_{f_L(N)} T = \frac{\log_N T}{\log_N f_L(N)} < R \Rightarrow \log_N T < \log_N f_L(N) * R$

$$\Rightarrow |\sqrt{\log_N T} * \sqrt{\log_N T}| = |\log_N f_L(N) * R|$$

$$\Rightarrow |\log_N T| = |\log_N f_L(N) * R|$$

$$\Rightarrow \left|\frac{\log_N T}{\log_N f_L(N)}\right| = |R| \Rightarrow |\log_{f_L(N)} T| = |R|.$$

Thus, based on R, any combination of these seven operations yields a result that is still equipotent to R.

It can be seen that the cardinality of the set described by N, given by $|f(N)|$, belongs to the set $\{|N|, |R|\}$ and does not admit other values. That is, it is impossible to describe a set S in the language of natural numbers such that $|N| < |S| < |R|$. Just as it is impossible to paint a blue sea with a red brush, even though a blue sea may exist.

5 Computations of Real-Related Sets

$\forall\, k\, (k \in N)$, $R < R^k < R^R$, $R < k^R < R^R$.
If $|R| = |R^R|$, then $|R| = |R^k| = |k^R| = |R^R|$.

$$\log_N R^R = R * \log_N R.$$

Since $|R| = |\log_N R| = \left|\log_N^{\log_N R}\right|$ and $\log_N^{\log_N R} < \sqrt{\log_N R} < \log_N R < R$, we have

$$|R| = \left|\sqrt{\log_N R}\right| \quad \text{and} \quad |\log_N R| = \left|\sqrt{\log_N R}\right|.$$

Thus $|R * \log_N R| = \left|\sqrt{\log_N R} * \sqrt{\log_N R}\right| = |\log_N R|$, we have

$$\left|N^{R * \log_N R}\right| = \left|N^{\log_N R}\right|.$$

Therefore $|R^R| = |R|$.
This can be proved by induction:

$$|R| = |f_L(R)| = \left|R^{R^{\cdots^R}}\right|,\ k\ R\text{s},\ k \in N.$$

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